

Lab 19*

Cancer Mutation Mini-Project

Instructions

Save this document to your computer and open it in a PDF viewer such as Preview (available on every mac) or Adobe Acrobat Reader ([free for PC and Linux](#)). Be sure to add your name and UC San Diego personal identification number (PID) and email below before answering all questions in the space provided.

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Background:

To identify somatic mutations in a tumor, DNA from the tumor is sequenced and compared to DNA from normal tissue in the same individual using variant calling algorithms.

Comparison of tumor sequences to those from normal tissue (rather than ‘the human genome’) is important to ensure that the detected differences are not germline mutations.

To identify which of the somatic mutations leads to the production of aberrant proteins, the location of the mutation in the genome is inspected to identify non-synonymous mutations (i.e. those that fall into protein coding regions and change the encoded amino acid).

As you go through this mini-project please remember to:

- Download the PDF version of this lab sheet (as noted above).
- Type all your answers directly in the space provided below each question.
- Save and upload your completed PDF to [gradescope](#).

Good luck!

*<http://thegrantlab.org/teaching/>

Questions:

Visit the following [webpage](#) and download your student specific sequences for the **Spring 2026** quarter. These sequences resulted from an NGS analysis of patient healthy and tumor tissue.

N.B. Note that these sequence are unique for you and you must download your sequences and use them to answer the following questions in the space provided.

Q1. [1pt] What protein do these sequences correspond to? (Give both full gene/protein name and official symbol).

Full: Phosphatidylinositol 4,5-bisphosphate 3-kinase catalytic subunit alpha isoform [Homo sapiens]
Official symbol: PIK3CA (PIK3CA gene encodes p110alpha catalytic subunit of PI3Kalpha enzyme)

Q2. [6pts] What are the tumor specific mutations in this particular case (e.g. A130V)?

The tumor specific mutations are: S553V, K573E, V616R, and E642Y.

Q3. [1pts] Do your mutations cluster to any particular domain and if so give the name and PFAM id of this domain? Alternately note whether your protein is single domain and provide it's PFAM id/accession and name (e.g. PF00613 and PI3Ka).

The overall, 1068 residue sequence has 5 PFAM domains (see pgs. 6, 7), but the mutations cluster to the PI3Ka domain (full: Phosphoinositide 3-kinase family, accessory domain) with id PF00613.

Q4. [2pts] Using the [NCI-GDC](#) list the observed top 2 missense mutations in this protein (amino acid substitutions)?

The top 2 missense mutations in this protein (see pg. 7) are E545K (DNA base substitution is chr3:g.179218303G>A) and H1047R (DNA base substitution is chr3:g.179234297A>G).

Q5. [2pts] What two TCGA projects have the most cases affected by mutations of this gene? (Give the TCGA “code” and “Project Name” for example “TCGA-BRCA” and “Breast Invasive Carcinoma”).

By percentage of cases affected by PIK3CA mutations (see pg. 8), the top two TCGA projects are:
(1) TCGA-UCEC, Uterine Corpus Endometrial Carcinoma (49.41% of cases affected)
(2) TCGA-UCS, Uterine Carcinosarcoma (35.09% of cases affected)
* Note: By number (not percent) of affected cases, top 2 projects are TCGA-BRCA (Breast Invasive Carcinoma; 331 cases) and TCGA-UCEC (Uterine Corpus Endometrial Carcinoma; 253 cases)

Q6. [3pts] List one RCSB PDB identifier with 100% identity to the wt_healthy sequence and detail the percent coverage of your query sequence for this known structure? Alternately, provide the most similar in sequence PDB structure along with its percent identity, coverage and E-value. Does this structure “cover” (i.e. include or span the amino acid residue positions) of your previously identified tumor specific mutations?

PDB ID: 2RD0_1 (Structure of a human p110alpha/p85alpha complex, Entity 1; see pg. 8)
Sequence percent identity: 100% ; Percent coverage of query sequence: 100% ; E-value: 0
Since this structure covers 100% of the query sequence, it spans the amino acid residues of the previously identified tumor specific mutations (positions 553, 573, 616, and 642 in wt_healthy sequence).

Optional Extension:

The following 3 questions are not required for this lab session but are here for motivated students to see how “druggable hot-spots” near these mutation sites might be identified:

Q7. [10pts] Using [AlphaFold notebook](#) generate a structural model using the default parameters for your **mutant** sequence.

Note that this can take some time depending upon your sequence length. If your model is taking many hours to generate or your input sequence yields a “too many amino acids” (i.e. length) error you can focus on the main PFAM domain of interest (your answer to Q3 above).

Once complete save the resulting PDB format file for your records and use [Mol-star](#) (or your favorite molecular viewer) to render a molecular figure. In this figure please clearly show your mutant amino acid **side chains as spacefill** and the protein as **cartoon colored by local alpha fold pLDDT quality score**. This score is contained in the B-factor column of your PDB downloaded file. **Send this image to Barry for some bonus points.**

Emailed to Prof. Grant and Fish. Also attached on page 9.

Q8. [2pts] Considering only your mutations in high quality structure regions (with a pLDDT score > 70) are any of the mutations on the surface of the protein and hence have a potential to interfere with protein-protein interaction events? List these mutations below (e.g. A130V)

All four mutated residues have pLDDT scores above 70: 91.25 (S553V), 95.56 (K573E), 95.44 (V616R), and 86.88 (E642Y). S553V and K573E are exposed and on the surface of the protein (see polymer colored by Accessible Surface Area on pg. 10), whereas V616R and E642Y are buried. Thus, S553V and K573E are most likely to interfere with protein-protein interaction events.

Q9. [5pts] Please comment on how useful and/or reliable you think your AlphaFold structural model is for your entire sequence and the main domain where your mutations lie? You may wish to compare your model to the PDB structure you found in Q6.

I think my AlphaFold model is reliable for the overall sequence, as well as for the main domain (PIK3a) where my mutations are found. The median pLDDT quality score is 93.75 for the entire sequence and 95.41 for the PIK3a domain, with the high scores indicating high confidence in the prediction. This is visually apparent from the largely-blue, pLDDT-colored Mol* model (page 9; the few low-confidence, orange/yellow regions mostly correspond to loops, rather than to regular secondary structures) and the pLDDT line plot (page 12, top) where the annotated PIK3a domain range has consistently high scoring. Furthermore, the PAE score plot for the AlphaFold model is mostly dark blue for the PIK3a domain (page 12, bottom; resno 520-703), indicating low PAE scores and that the domain where the mutations lie has low estimated positional error. Upon using Mol* to superpose the Q6 PDB structure (2RD0_1) and the AlphaFold model (see page 11), the two appear to align well – particularly at alpha folds/beta helices, but even at loops.

Q10. [10pts] Visit the [FTMap online server](#) and sign-up for an account. Once logged in upload your PDB structure model, provide a job name and submit. Note that that these jobs can take some time (over a day) to run if there are multiple runs scheduled ahead of you.

FTMap aims to identify potential **binding hot spots** on the surface of your protein that can bind with high affinity to ligands, drugs, or other proteins. The server does this by docking

a set of small chemical probes onto the protein surface. The server then clusters the probes based on their spatial proximity. Each cluster represents a potential binding site, with larger clusters generally indicating stronger or more likely binding sites.

NB. Note that in advanced options you can also run in **PPI Mode** with the goal of detecting binding hot spots for protein-protein interactions using an alternative set of parameters. You might wish to try this as a second calculation.

Are any of the identified “hot spots” near your cancer specific mutation sites or the most commonly mutated sites from the NCI-GDC? If so which mutation site(s)?

I was unable to access FTMap: “ This site can ’ t be reached. ftmap.bu.edutook too long to respond. ” I attempted troubleshooting and trying again the next day, but the site appears to be down

- End of Lab -

Q3: I inputted the protein sequence into PFAM/InterPro sequence search, which outputted 5 PFAM domain matches (“blue bars” in top screenshot, and listed out in bottom screenshot). The Q2 mutations cluster specifically to the PI3Ka domain (residues 520-703), PF00613.


InterPro




Classification of protein families

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[🏠 / Result / InterProScan / Iprscan5-R20260609-015315-0620-22171566-P1m / Entry / Pfam](#)

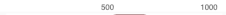
InterProScan Search Result InterProScan protein

Overview
Entries 14
Sequence

This protein matches these entries:

1 - 5 of 5 entries matching Pfam[®]


H

Accession	Short Name	Name	Source Database	Matches
PF02192	PI3K_p85B	PI3-kinase family, p85-binding domain	Pfam	
PF00613	PI3Ka	Phosphoinositide 3-kinase family, accessory domain (PIK domain)	Pfam	
PF00794	PI3K_rbd	PI3-kinase family, ras-binding domain	Pfam	
PF00792	PI3K_C2	Phosphoinositide 3-kinase C2	Pfam	
PF00454	PI3_PI4_kinase	Phosphatidylinositol 3- and 4-kinase	Pfam	

Rows per page: 20
Previous 1 Next

Q3: Same output as above (5 PFAM matches, with PI3Ka specific to mutations), using HMMER search instead

Browse Phmmer Results

[Score](#)[Taxonomy](#)[Domain](#)[Download](#)

Go to Batch job summary

PI3Ka (PF00613)

Description: Phosphoinositide 3-kinase family, accessory domain (PIK domain)

Coordinates: 519 - 704 (alignment region 520 - 703)

Model Match: 1 185

Source: pfam

PI3K_rbd

PI3K_C2

PI3Ka

PI3_P14_kinase

Distribution of Significant Hits

more significant

BacteriaArchaeaEukaryotaVirusesUnclassified SequencesOther Sequences

Search Details

Customise Table

Target	Description	Species	E-value
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Q4: Top 10 observed mutations (all missense substitutions) in PIK3CA-encoded protein.

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Obtaining Access to Controlled Data

Q e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G

GENE • PIK3CA

MOST FREQUENT SOMATIC MUTATIONS

Save/Edit Mutation Set

TSV

TOTAL OF 436 SOMATIC MUTATIONS

Q

Search

☐	DNA Change	Protein Change	Type	Consequences	# Affected Cases in PIK3CA	# Affected Cases Across the GDC	Impact
☐	chr3:g.179218303G>A	PIK3CA E545K	Substitution	Missense	+ 332 / 5,791 (5.73%)	332 / 18,289 (1.82%)	MO DH PR
☐	chr3:g.179234297A>G	PIK3CA H1047R	Substitution	Missense	+ 303 / 5,791 (5.23%)	303 / 18,289 (1.66%)	MO TO BE
☐	chr3:g.179218294G>A	PIK3CA E542K	Substitution	Missense	+ 204 / 5,791 (3.52%)	204 / 18,289 (1.12%)	MO DH PR
☐	chr3:g.179199088G>A	PIK3CA R88Q	Substitution	Missense	+ 85 / 5,791 (1.47%)	85 / 18,289 (0.46%)	MO TO PR
☐	chr3:g.179234297A>T	PIK3CA H1047L	Substitution	Missense	+ 49 / 5,791 (0.85%)	49 / 18,289 (0.27%)	MO TO BE
☐	chr3:g.179203765T>A	PIK3CA N345K	Substitution	Missense	+ 41 / 5,791 (0.71%)	41 / 18,289 (0.22%)	MO DH PR
☐	chr3:g.179221146G>A	PIK3CA E726K	Substitution	Missense	+ 37 / 5,791 (0.64%)	37 / 18,289 (0.20%)	MO TO BE
☐	chr3:g.179210192T>C	PIK3CA C420R	Substitution	Missense	+ 34 / 5,791 (0.59%)	34 / 18,289 (0.19%)	MO TO PO
☐	chr3:g.179199690G>A	PIK3CA G118D	Substitution	Missense	+ 30 / 5,791 (0.52%)	30 / 18,289 (0.16%)	MO TO PO
☐	chr3:g.179210291G>A	PIK3CA E453K	Substitution	Missense	+ 25 / 5,791 (0.43%)	25 / 18,289 (0.14%)	MO TO PO

Show

10

Entries

Showing 1 - 10 of 436 somatic mutations

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1

2

3

4

5

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44

>

>>

Q5: Top 10 TCGA projects affected by mutations of PIK3CA gene

portal.gdc.cancer.gov

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Obtaining Access

GENE • PIK3CA

JSON TSV

TOTAL OF 44 PROJECTS

Project	Disease Type	Primary Site	# SSM Affected Cases	# CNV Amplifications	# CNV Homozygous Deletions	# Mutations
TCGA-UCEC	4 Disease Types	2 Primary Sites	253 / 512 (49.41%)	51 / 540 (9.44%)	0 / 540 (0.00%)	151
TCGA-UCS	2 Disease Types	2 Primary Sites	20 / 57 (35.09%)	15 / 55 (27.27%)	0 / 55 (0.00%)	15
TCGA-BRCA	9 Disease Types	Breast	331 / 969 (34.16%)	131 / 1,084 (12.08%)	0 / 1,084 (0.00%)	76
CML-MBC	Ductal and Lobular Neoplasms	Breast	52 / 174 (29.89%)	0 / 0 (0.00%)	0 / 0 (0.00%)	17
TCGA-COAD	4 Disease Types	2 Primary Sites	125 / 428 (29.21%)	4 / 454 (0.88%)	0 / 454 (0.00%)	59
CGIC-HTMCP-CC	4 Disease Types	Cervix uteri	24 / 85 (28.24%)	44 / 122 (36.07%)	0 / 122 (0.00%)	11
TCGA-CESC	4 Disease Types	2 Primary Sites	81 / 287 (28.22%)	124 / 299 (41.47%)	0 / 299 (0.00%)	31
TCGA-BLCA	4 Disease Types	Bladder	86 / 408 (21.08%)	50 / 402 (12.44%)	0 / 402 (0.00%)	42
CPTAC-2	5 Disease Types	6 Primary Sites	64 / 328 (19.51%)	0 / 0 (0.00%)	0 / 0 (0.00%)	40
TCGA-HNSC	Squamous Cell Neoplasms	13 Primary Sites	86 / 509 (16.90%)	176 / 519 (33.91%)	0 / 519 (0.00%)	35

Show 10 Entries

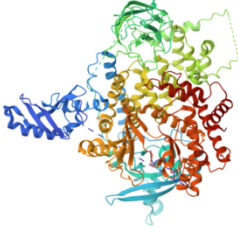
Showing 1 - 10 of 44 projects

<< < 1 2 3 4 5 > >>

Q6: PDB ID 2RD0

Download Multiple Files: All Selected

Download File View File Download Alignment



Explore in 3D

2RD0: Entity 1

Structure of a human p110alpha/p85alpha complex

Huang, C., Gabelli, S.B., Amzel, L.M.

(2007) Science 318: 1744-1748

Released2007-12-25

MethodX-RAY DIFFRACTION 3.05 Å

Chain IDA

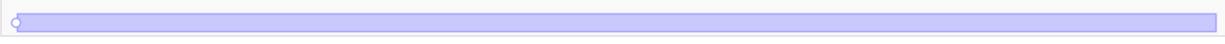
OrganismHomo sapiens

MacromoleculePhosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha isoform

Sequence MatchSequence Identity: 100%, E-Value: 0, Region: 29-1096

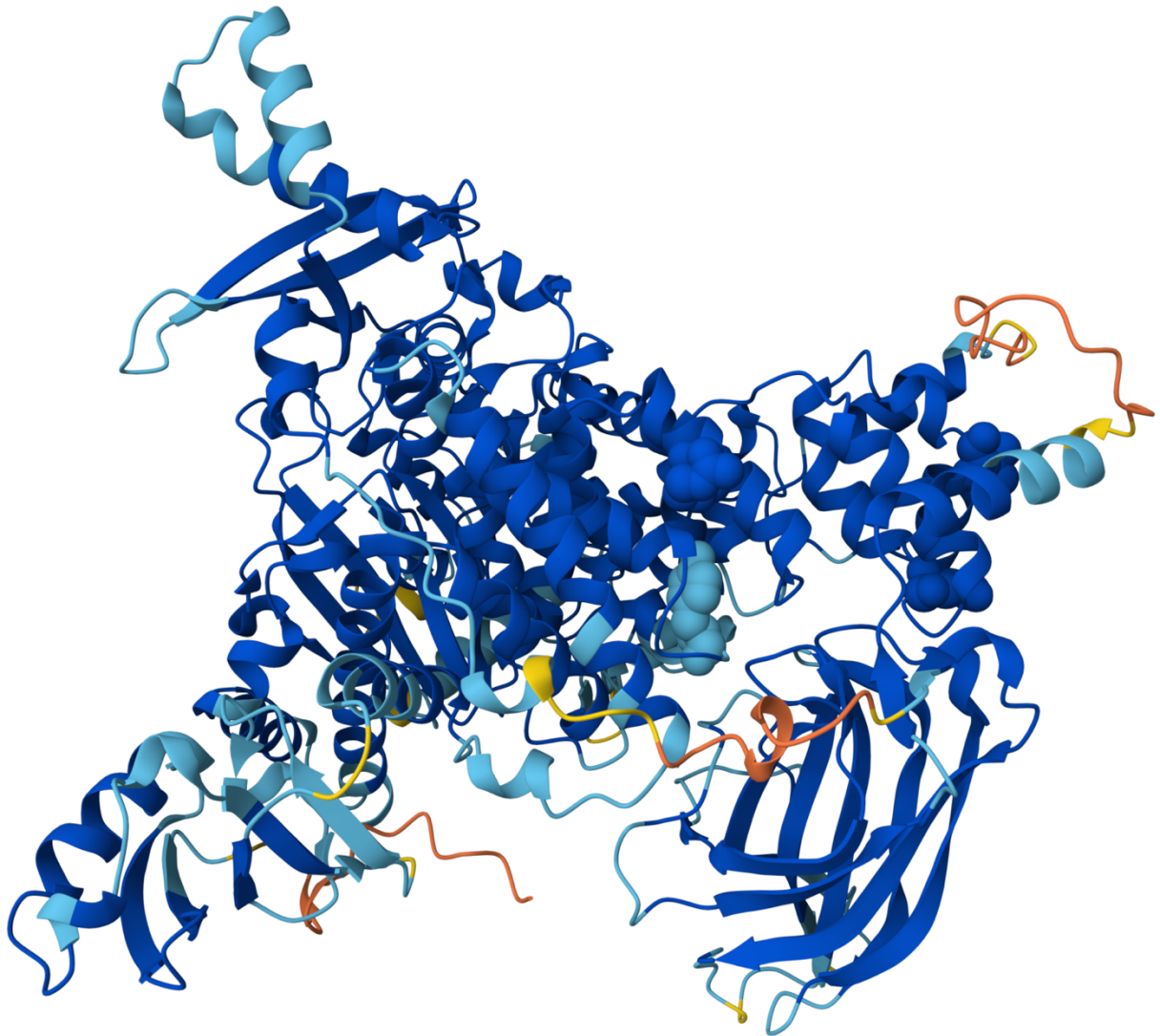
01002003004005006007008009001,000

QUERY2RD0_1

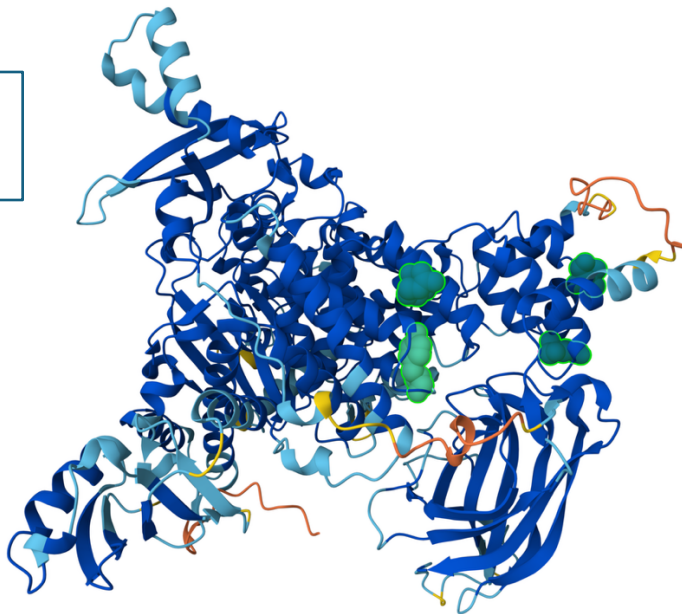


Note: 2RD0 is the PI3Kalpha dimer, with (1) the PIK3CA-encoded p110alpha catalytic subunit (2RD0_1; shown above) and (2) the p85alpha regulatory subunit (2RD0_2). The wt_healthy query sequence for PIK3CA-encoded p110alpha is covered by residues 29-1096 of 2RD0_1.

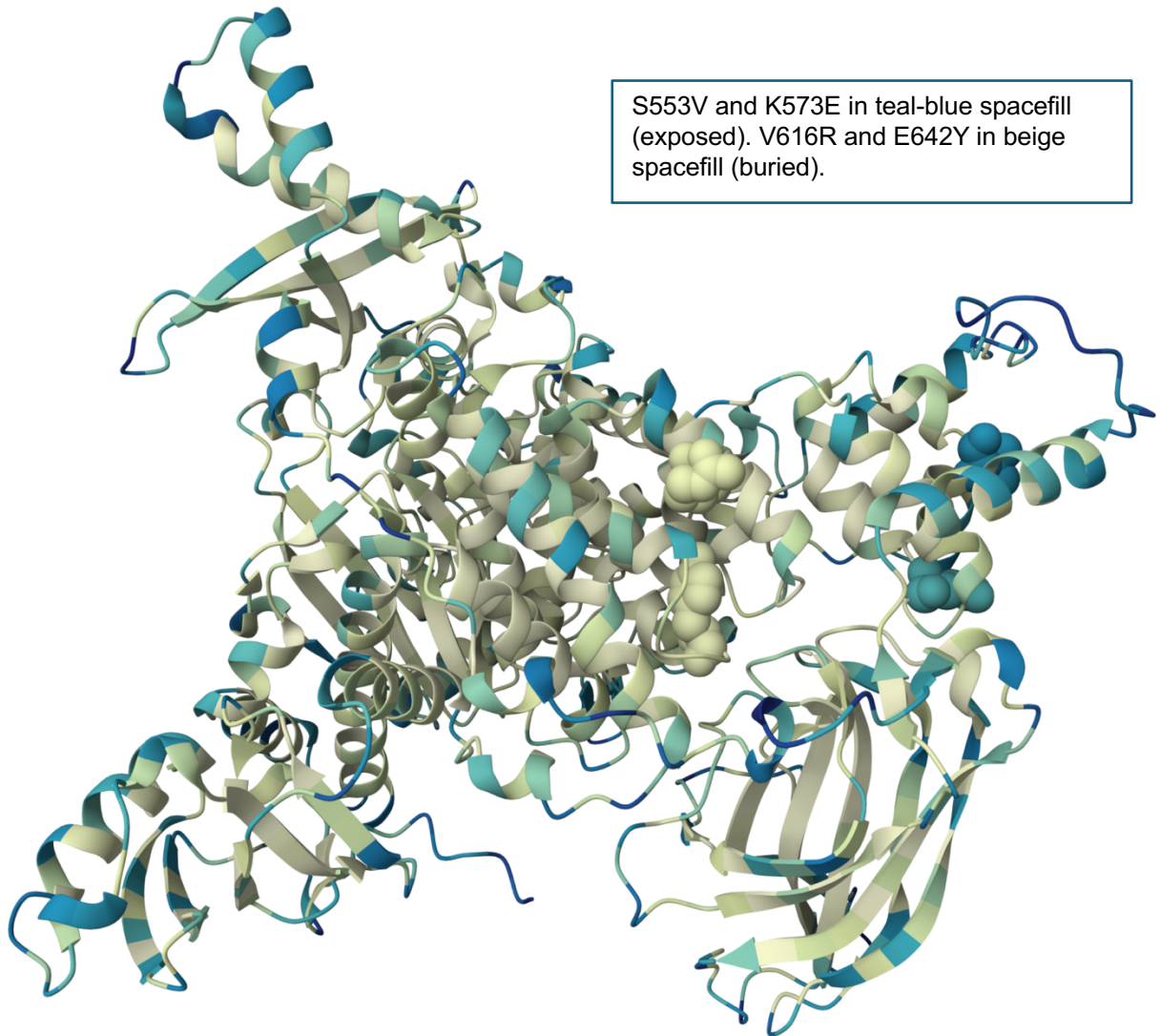
Q7: AlphaFold structural model of mutant sequence, colored by pLDDT quality score. Mutant amino acid side chains are highlighted as spacefill.



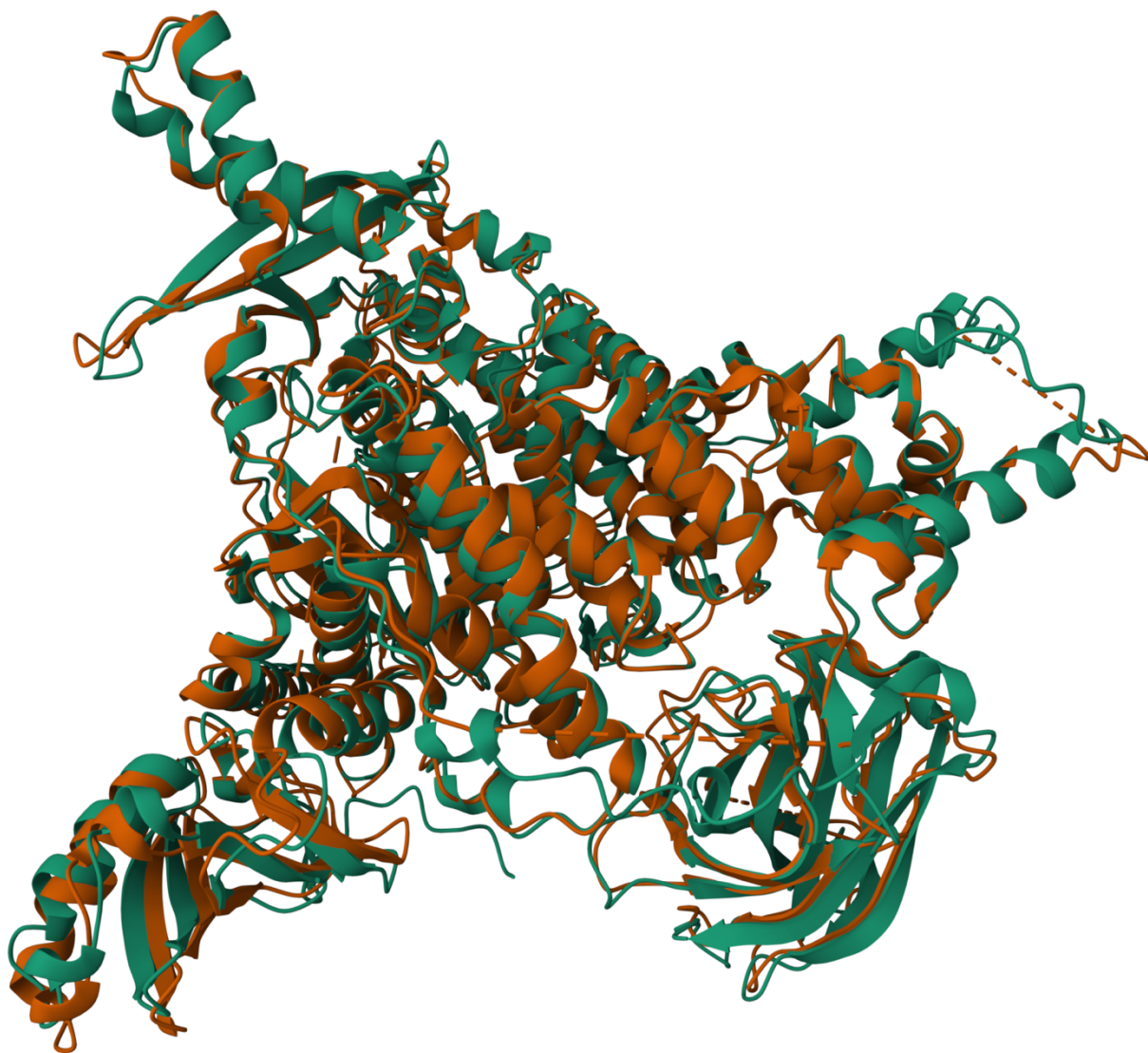
Another image, with
residues selected for
easier locating.



Q8: Same structure shown above, now colored by “Accessible Surface Area” (beige corresponds to buried residues, whereas blue corresponds to exposed residues)

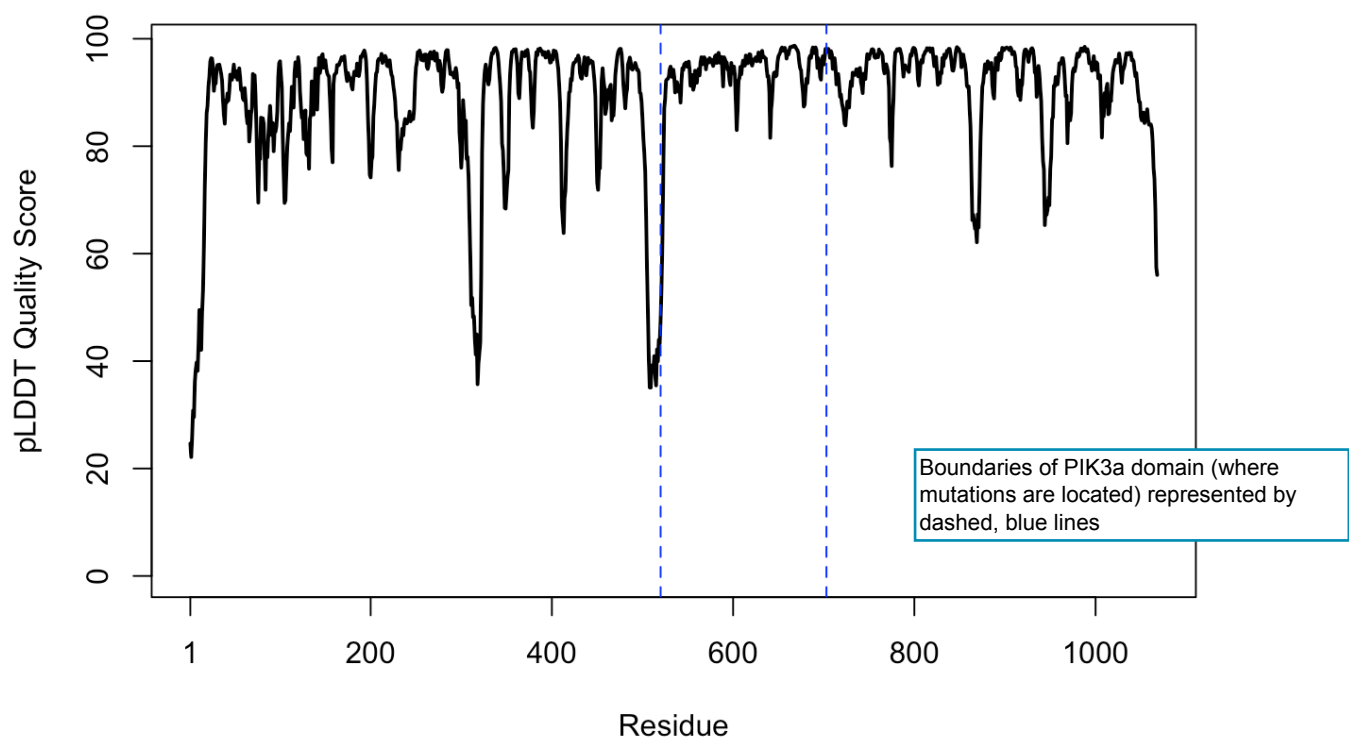


Q9: Superposition of AlphaFold structure prediction (green) and PDB 2RD0_1 (orange).



Q10: pLDDT and PAE plots for AlphaFold model

Plot of pLDDT Values for AlphaFold Rank 001 Model



PAE Score Plot for AlphaFold Rank 001 Model

